JAVA PROGRAMMING

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C02-AT1-SCENARIOBASED QUESTIONS

Q1 **Student Registration System** 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

```
import java.util.*;

public class Student {

    public static void main(String[] args) {

        HashSet<String> students = new HashSet<>();
        students.add("101-Alice");
        students.add("102-Bob");
        students.add("103-Charlie");

        HashMap<String, ArrayList<String>> courseMap = new HashMap<>();

        courseMap.put("101-Alice", new ArrayList<>(Arrays.asList("Java"
        courseMap.put("102-Bob", new ArrayList<>(Arrays.asList("Python"
        courseMap.put("103-Charlie", new ArrayList<>(Arrays.asList("C++

        Iterator<String> it = students.iterator();

        while (it.hasNext()) {
            String s = it.next();
            System.out.println("Student: " + s);
            System.out.println("Courses: " + courseMap.get(s));
        }
    }
}
```

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

• Add items

• Remove an item

• Display all items

Your Code

```java
import java.util.*;

class Main {
    public static void main(String[] args) {

        ArrayList<String> cart = new ArrayList<String>();

        cart.add("Laptop");
        cart.add("Mouse");
        cart.add("Keyboard");
        cart.add("Headphones");

        System.out.println("Items in Cart:");

        for (String item : cart) {
            if (!item.equals("Mouse")) {
                System.out.println(item);
            }
        }
    }
}
```

Input

stdin input

Output

```
Items in Cart:
Laptop
Keyboard
Headphones
```

Submitted

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

• Insert book details

• Retrieve a book

• Display all books

Your Code

```java
import java.util.*;

class Main {
    public static void main(String[] args) {

        HashMap<String, String> books = new HashMap<String, String>();

        // Insert book details
        books.put("978001", "Java Programming");
        books.put("978002", "Data Structures");
        books.put("978003", "Database Systems");

        // Retrieve a book
        System.out.println("Retrieved Book:");
        System.out.println(books.get("978002"));

        // Display all books
        System.out.println("\nAll Books:");
        for (Map.Entry<String, String> entry : books.entrySet()) {
            System.out.println("ISBN: " + entry.getKey() + " Book: " +
        }
    }
}
```

Input

```
stdin input
```

Output

```
Retrieved Book:
Data Structures

All Books:
ISBN: 978001 Book: Java
Programming
```

Submitted

Q4 **Employee Data Processing** 10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

• Store employee objects

• Display employees with salary > 50,000

```java
import java.util.*;

class Main {
    public static void main(String[] args) {

        List<String> employees = new ArrayList<String>();
        List<Integer> salaries = new ArrayList<Integer>();

        employees.add("Alice");
        salaries.add(45000);

        employees.add("Bob");
        salaries.add(60000);

        employees.add("Charlie");
        salaries.add(75000);

        System.out.println("Employees with salary > 50000");

        for (int i = 0; i < employees.size(); i++) {
            if (salaries.get(i) > 50000) {
                System.out.println(employees.get(i) + " " + salaries.ge
            }
        }
    }
}
```

```
stdin input
```

Output

```
Employees with salary > 50000
Bob 60000
Charlie 75000
```

Q5 **Online Voting System** 10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

• Record votes

• Prevent duplicate voting

• Display results

```java
import java.util.*;

class Main {
    public static void main(String[] args) {

        HashSet<String> voters = new HashSet<String>();
        HashMap<String, Integer> votes = new HashMap<String, Integer>()

        // Record votes
        if (voters.add("Voter1")) {
            votes.put("CandidateA", votes.getOrDefault("CandidateA", 0)
        }

        if (voters.add("Voter2")) {
            votes.put("CandidateB", votes.getOrDefault("CandidateB", 0)
        }

        // Duplicate vote (will not be counted)
        if (voters.add("Voter1")) {
            votes.put("CandidateA", votes.getOrDefault("CandidateA", 0)
        }

        // Display results
        System.out.println("Voting Results:");
        for (String candidate : votes.keySet()) {
            System.out.println(candidate + " : " + votes.get(candidate)
        }
    }
}
```

stdin input

Output

```
Voting Results:
CandidateA : 1
CandidateB : 1
```